

The Hippocampus

The hippocampus in spatial navigation

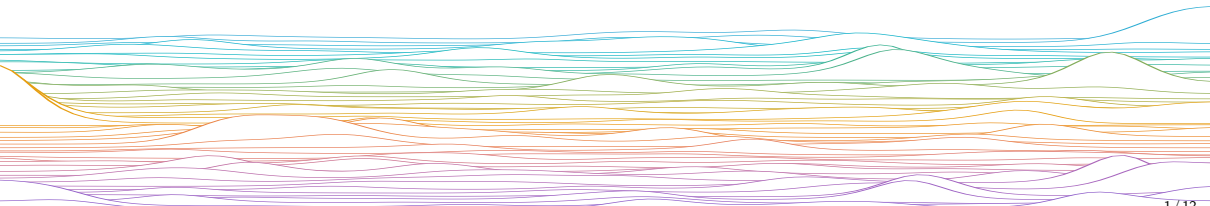
Computational Neuroscience

University of Bristol

mrule

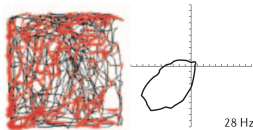
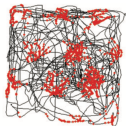
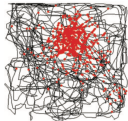
Learning outcomes:

- ▶ Be able to discuss tuning of place/grid/memory/head-direction cells
- ▶ Report the distinct roles of place/memory cells and grid cells in navigation



Hippocampus and spatial navigation

Neurons in hippocampus & surrounding regions respond to the spatial environment



Place cells

- ▶ CA3 and CA1
- ▶ Active only when an animal is one particular location

Grid cells

- ▶ Entorhinal cortex
- ▶ Active when the animal is in a set of locations arranged in a hexagonal grid

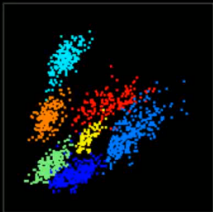
Head-direction cells

- ▶ Subiculum & throughout
- ▶ Active only when the animal is facing a certain direction.

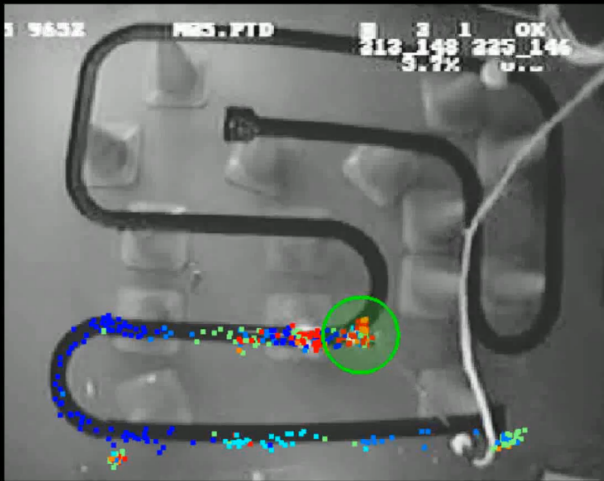
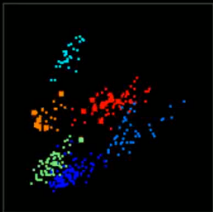
cell activity

behavior

overall



ongoing



0:25 / 1:14



[Video: Hippocampal place cells recorded in the Wilson lab at MIT](#)

Memory or navigation?

In both memory and spatial navigation, hippocampus is thought to perform two classes of computation:

Pattern separation/completion

Place cells → **memory cells**

Store/encode/decode a

I'm here, this is happening state

(call it '**y**')

Pattern separation:

- ▶ Dentate Gyrus (DG)

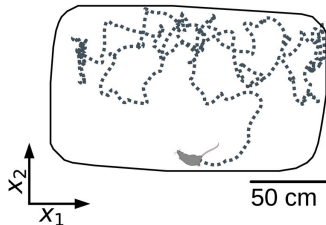
Pattern Completion

- ▶ Recurrent CA3 attractor net

Path integration

Keep & update internal estimate of location by integrating velocity

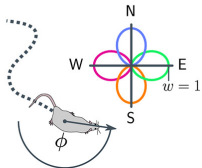
Grid-like cells make it easier to integrate changes $\Delta \mathbf{y} \rightarrow$ into estimates $\hat{\mathbf{y}}$



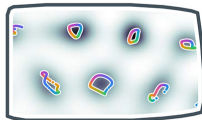
Just for fun: A grid $\otimes$ heading (?) cell from the Krupic lab

(a) Head-direction weights

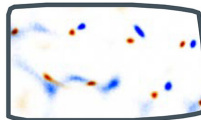
$$w = \max[0, \cos(\phi - \phi_0)]^2$$



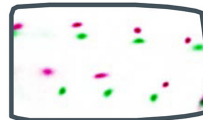
(c) Tracked peaks



(d) Northwards (blue) vs southwards (orange)

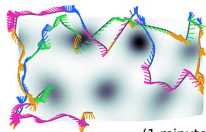


Eastwards (green) vs westwards (pink)



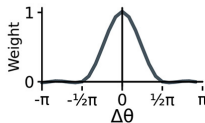
50 cm

(b) Head direction



(1 minute excerpt)

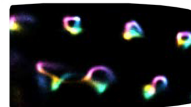
(e) Head-direction kernel



(f) Joint rate map



(g) Joint peak density



E S W N

[Rule et al. 2023 Hippocampus](#)

The specifics of grid cell path integration and the successor representational theory are extremely cool. However, this would lead to a few too many further facts and models to memorize for the exam

We will discuss navigation again in context of continuous attractors in the next lecture.

end

